E3.4.1 Renewable and non-renewable energy sources

Renewable sources replenish on timescales comparable to human use, including solar, wind and hydropower. Non-renewable resources are finite and not quickly replaced, including coal, oil, natural gas and conventional nuclear fuels. Replenishment, not carbon emissions during generation, determines the classification. Nuclear power still uses finite fuel despite comparatively low operating carbon emissions.

E3.4.2 Total binding energy, binding energy per nucleon and reaction energy

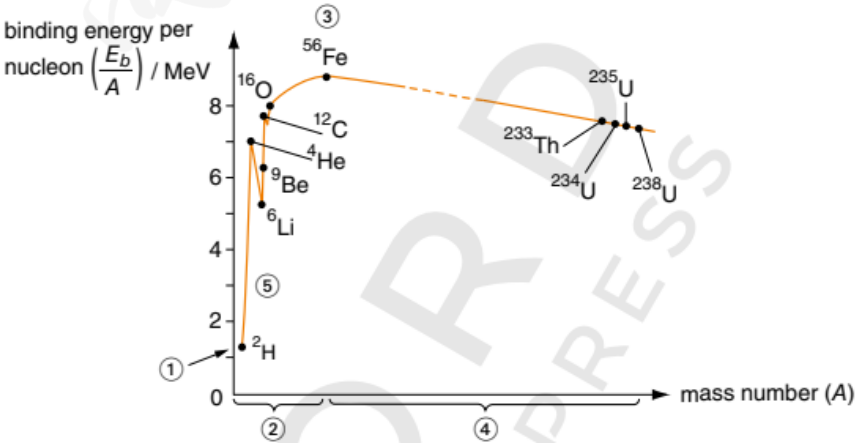
Binding energy E_b is the energy needed to separate a nucleus completely into free nucleons. The same energy is released when they form the nucleus. Their combined free mass exceeds nuclear mass by the mass defect Δm. Treat electron masses consistently when using atomic masses.

Binding energy per nucleon is E_b/A, total binding energy divided by nucleon number. A larger value generally means more tightly bound nucleons and greater nuclear stability. The textbook curve is high near iron. Fusion of light nuclei or fission of heavy nuclei can form products with greater binding energy per nucleon and release energy.

Multiply each value read from the curve by A to obtain total binding energy. Released energy equals total product binding energy minus total reactant binding energy, including the number of each nucleus in the equation. Free nucleons have zero binding energy.

$$\frac{E_b}{\mathrm{MeV}} \approx 931 \frac{\Delta m}{\mathrm{u}}$$

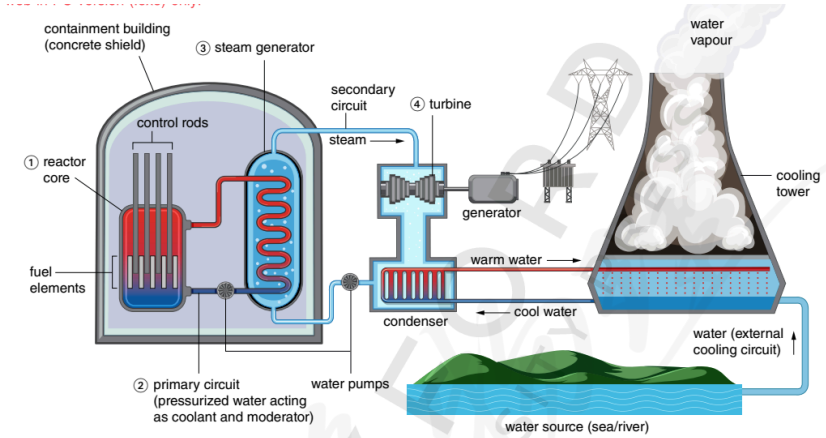
E_b: binding energy; Δm: mass of separated nucleons minus nuclear mass. 1 u corresponds to approximately 931 MeV/c².



Binding energy per nucleon is high near iron. Light-nucleus fusion and heavy-nucleus fission can form more tightly bound products.

E3.4.3 Components and circuits of a pressurised water reactor

Fission in the fuel releases energy. Control rods absorb neutrons to regulate the chain reaction. High-pressure water in the primary circuit removes heat and also acts as a moderator, slowing neutrons. Pressure keeps this water liquid at high temperature. The steam generator transfers heat to a separate secondary circuit. Its steam drives the turbine and generator, then gives up heat to an external cooling circuit in the condenser. The steam condenses to water for reuse. The circuits exchange heat while keeping their water separate.



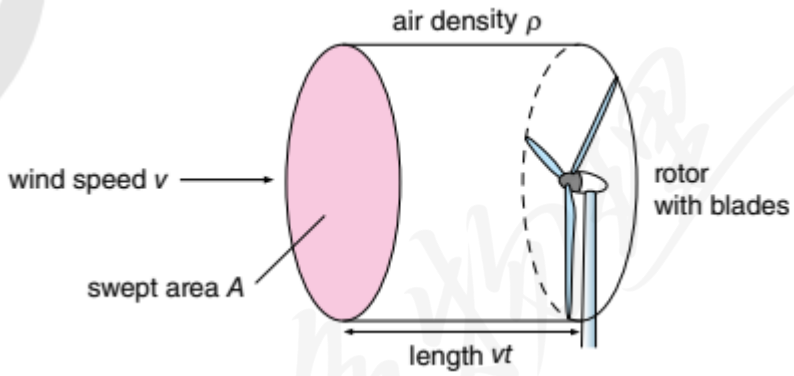
The primary, secondary and external cooling circuits of a pressurised water reactor.

E3.4.4 Wind-turbine power

Wind carries kinetic energy through the rotor's swept area. Available kinetic-energy flux is proportional to air density, swept area and wind speed cubed. Actual output also includes efficiency. Swept area is the circle covered by rotating blades, not their material area. With other conditions fixed, doubling speed increases this model's power eightfold.

$$P = \frac{1}{2}\eta\rho A v^3$$

Wind turbine output P (W); η: fractional efficiency; air density ρ (kg m^{-3}); swept rotor area A (m^2); wind speed v (m s^{-1}). Uniform wind through A.



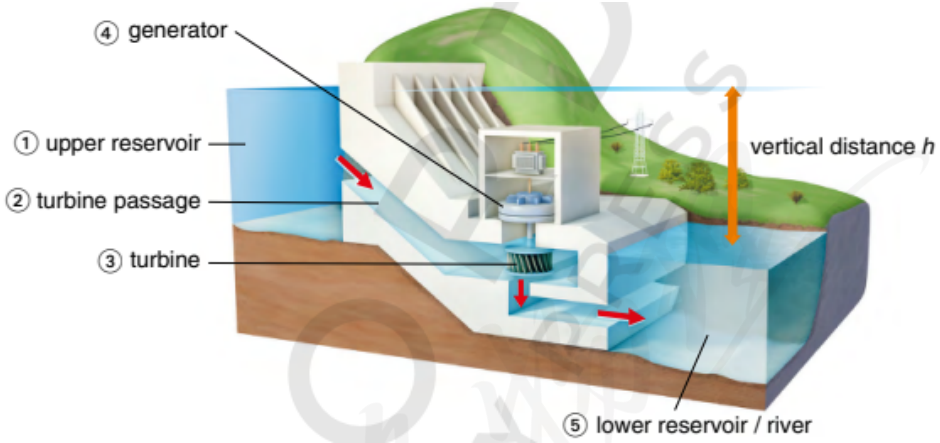
The air cylinder crossing the swept rotor area in a time interval has length equal to wind speed times time.

E3.4.5 Hydroelectric power

Falling water loses gravitational potential energy, part of which turbines and generators convert to electricity. Output power depends on mass flow per second, vertical head and efficiency. Convert volume flow rate to mass flow rate using water density. Head is the vertical height difference, not total pipe length.

$$P = \eta \frac{mgh}{t}$$

Hydroelectric output P (W); η: fractional efficiency; m/t: mass flow rate (kg s^{-1}); g: m s^{-2}; h: vertical head (m).



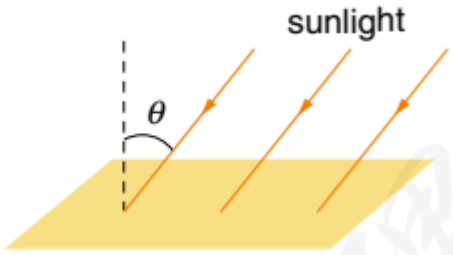
Water flows from the upper to lower reservoir, converting gravitational potential energy to electricity through a turbine and generator.

E3.4.6 The solar constant

The solar constant is radiant power received per unit area perpendicular to sunlight, outside Earth's atmosphere at the mean Earth-Sun distance. It is a power density, not the Sun's total power or a value received everywhere at ground level. Atmospheric absorption, clouds and oblique incidence alter the actual received power.

E3.4.7 Solar power and projected area

For sunlight intensity I measured perpendicular to its rays, a panel of area A intercepts projected area $A\cos\theta$, where θ is the angle from the panel normal. Incident power is $P = IA\cos\theta$, greatest at normal incidence when $\theta = 0$. If the angle is measured from the panel surface, use sine instead. If actual power per panel area is already given, multiply by A without another angle factor. Electrical output also includes conversion efficiency.

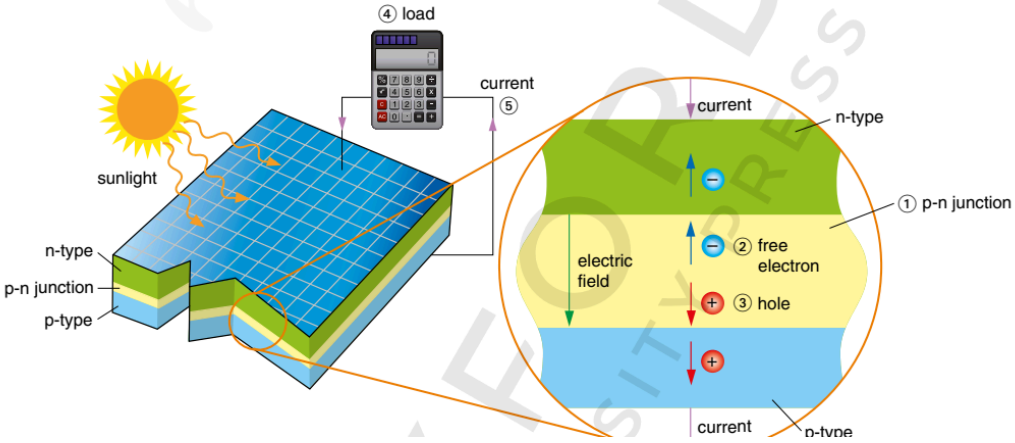


θ is measured between the light rays and the panel normal.

E3.4.8 The photovoltaic effect in a solar cell

A solar cell uses a p–n junction to convert light into electrical energy. Electrons are the majority carriers on the n-side and holes on the p-side. A hole is a missing electron that acts as a positive carrier.

Light generates electrons and holes near the junction. Its electric field separates them, moving electrons towards the n-side and holes towards the p-side. With a load connected, electrons flow through the external circuit; conventional current is opposite.



Electric field, electron motion and external conventional-current directions in a photovoltaic cell.

E3.4.9 Anti-reflection films on solar panels

A transparent anti-reflection coating reduces reflected light, allowing more incident light into the cell for absorption and energy conversion. Reduced reflection also makes panels appear darker.

E3.4.10 The greenhouse effect

Shorter-wavelength solar radiation passes through the atmosphere and warms the surface. The surface emits longer-wavelength infrared radiation. Greenhouse gases absorb and re-emit some in all directions, reducing direct surface energy loss to space and raising equilibrium temperature. The main mechanism is radiative exchange, not simply sealing air inside a glass-like enclosure.

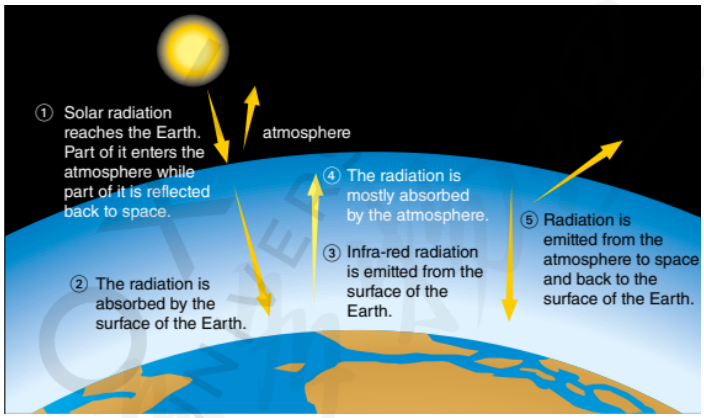


Fig 4.2b Greenhouse effect.

Short-wave sunlight and long-wave infrared radiation from Earth's surface follow different paths through the atmosphere.